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(57) Abstract :
The present invention discloses a modular thermoelectric energy harvesting system and method for recovering waste heat from multiple high-temperature zones of vehicles and industrial equipment. The system comprises a plurality of thermoelectric generator (TEG) modules configured in detachable arrays, a smart power conditioning unit for stabilizing fluctuating thermal inputs, an auxiliary energy storage module, and an IoT-enabled monitoring framework. The process involves converting distributed waste heat into electrical energy, conditioning the harvested power through adaptive regulation, storing surplus energy, and supplying regulated output to auxiliary vehicular subsystems such as headlights, dashboard electronics, and sensors. The modular plug-and-play architecture ensures retrofittable integration across diverse platforms including cars, two-wheelers, trucks, agricultural machinery, and stationary power units. The invention improves fuel efficiency by reducing alternator load while also enabling off-grid power support for portable lighting, emergency charging, and rural household electrification.

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